

SHIVANSH TYAGI

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AI/ML Engineer building production systems at the intersection of **satellite intelligence** and **LLM automation** - from optimizing pipelines to **ReAct agents** deployed at the **Ministry of Defence**. **First-author IEEE publication**. **SIH Finalist**.

Education

Raj Kumar Goel Institute of Technology (AKTU)

2022-2026

Bachelor of Technology in Computer Science and Engineering (CGPA: 7.8)

Ghaziabad, Uttar Pradesh

Skills

ML & AI: PyTorch, ONNX Runtime, Transformers, Hugging Face, Ollama, Llama.Cpp, LangGraph, LangChain, RAG

Geospatial: GDAL, GeoPandas, Shapely, QGIS, Fiona, Rasterio

Frameworks: FastAPI, Flask, Streamlit, React, Astro

Databases: PostgreSQL, MongoDB, Redis, Vector DBs (Qdrant, ChromaDB)

DevOps: Docker, Kubernetes, Git, AWS

Programming Languages: Python, Java, Javascript SQL

Experience

Ministry of Defence

Jan 2026 - Present

Machine Learning Intern

- Designed an end-to-end **automated intelligence system** combining AIS correlation, satellite imagery fusion, and a **Qwen-27B ReAct agent** (A100) to detect dark-activity and RF spoofing — replacing manual analyst workflows.
- Trained YOLO (78 mAP) and ResNet (83 accuracy) vessel models on real classified satellite imagery; both in active operational use in production.
- Architected an automated, Watchdog-driven pipeline to replace a fragmented two-script workflow, slashing processing of **70GB+** daily PLANET imagery from **3 hours to 25 minutes (6x speedup)** using tiled GDAL loading, tmpfs staging, and **GPU-based PyTorch** and **ONNX Runtime** optimizations.
- Engineered a custom dual-pool **GeoTIFF→MBTiles converter** (spawn+fork, tmpfs staging, tuned SQLite PRAGMAs) replacing **gdal2tiles** bottleneck for high-throughput national-scale satellite tile serving.
- Architected a unified, **highly concurrent AIS data orchestrator**, reducing per-file decoding time from **2 minutes to 30 seconds** and scaling to **30 parallel CPU workers**, resulting in a **120×** increase in data ingestion throughput.

Projects

EchoDB — Temporal Memory System for LLM Agents | [github](#) | [pypi](#)

- Architected and published** a persistent memory layer for LLM agents that enables long-term fact retention, structured knowledge extraction, and automated entity linking across conversation sessions.
- Engineered a **temporal graph framework** using **versioned chains** to manage state changes over time, ensuring agents retrieve the latest information while preserving a traceable historical context for updated facts.
- Implemented a **biologically-inspired memory** decay system and hybrid retrieval algorithm that prioritizes relevant context by balancing semantic similarity with access frequency and recency.
- Developed a **portable snapshot system (.umc)** storing graph + embeddings for instant restore without reprocessing

CLIP-Guided Image Optimization Engine - Dreams | [github](#)

- Engineered a custom **PyTorch pipeline** using **OpenCLIP (ViT-B/32)** to perform gradient ascent on image tensors, implementing custom differentiable **loss functions (Total Variation, Range Loss)** to maximize semantic alignment while preserving structural edge integrity.

Publications

Cognitively-Inspired Dual Memory for Long-Term Contextual Reasoning

IEEE, GTSS 2026

First Author — Presented — In Press

- Proposed D-mem, a dual-memory architecture separating factual (Semantic) and experiential (Episodic) knowledge via a bi-temporal knowledge graph (**Neo4j**) to enable **”as-of”** and **”since-when”** contextual reasoning.

Achievements

- Smart India Hackathon 2025 Finalist** — Led 6-member team; one of 2 teams to deliver a working product through the final round of India’s largest national govt. hackathon
- Winning the prize for Best use of MongoDB Atlas in HackHound 3.0.
- Achieved rank **8** out of 300+ teams in **IIT Delhi’s Tryst’25**
- Secured **1st** place in college Code Clash event **3 times**, awarded by **Striver**.
- Rated **3 star** at Codechef, **1127** on Codeforces.